MALARIA

Custom Study Summary with reference to National Health Portal (NHP) and Government of India malaria resources

Overview

Malaria is a potentially life-threatening parasitic disease caused by *Plasmodium* parasites. It is mainly transmitted through the bite of an infected female *Anopheles* mosquito. Malaria is preventable and treatable, but delayed diagnosis and treatment can lead to severe complications.

Key Facts

• Causative agent: Protozoan parasites of the genus *Plasmodium*.

• Important human malaria parasites include *P. vivax*, *P. falciparum*, *P. malariae* and *P. ovale*.

• Main vector: Infected female *Anopheles* mosquito.

• Incubation period: Symptoms commonly appear around 10–15 days after an infective mosquito bite.

• *P. falciparum* malaria is associated with a high risk of severe and life-threatening disease.

• Early diagnosis and complete treatment help prevent complications and reduce transmission.

Transmission

The main transmission cycle is:

Infected person → Anopheles mosquito → mosquito becomes infected → bite of another person → Malaria infection

• Malaria is primarily transmitted through the bite of an infected female Anopheles mosquito.

• The parasite undergoes stages of development in the human body and in the mosquito.

• Malaria does not normally spread through casual person-to-person contact.

• Transmission can also occur through infected blood in uncommon circumstances.

Clinical Features and Symptoms

Fever is the cardinal symptom of malaria. Symptoms can be non-specific and may resemble other febrile illnesses. Common features include:

• Fever, which may be intermittent or continuous

• Chills and rigors

• Headache

• Sweating

• Muscle and joint pain

• Fatigue and weakness

• Loss of appetite

• Nausea and vomiting

Severe and Complicated Malaria

Malaria, particularly *P. falciparum* infection, can rapidly progress to severe illness if treatment is delayed. Severe manifestations may include:

• Altered consciousness or cerebral malaria

• Repeated convulsions

• Severe anaemia

• Respiratory distress

• Jaundice

• Dark or bloody urine

• Abnormal bleeding

• Shock or multi-organ dysfunction

People at Higher Risk

• Infants and young children

• Pregnant women

• People with reduced immunity

• Travellers or people without previous immunity living in malaria-endemic areas

Diagnosis

Malaria should be confirmed through parasite-based diagnostic testing whenever possible. Common diagnostic methods include:

• Microscopic examination of peripheral blood smears

• Rapid Diagnostic Tests (RDTs)

• Clinical assessment together with laboratory confirmation

Testing helps distinguish malaria from other causes of fever and guides appropriate treatment.

Treatment and Management

Malaria requires appropriate antimalarial treatment. The choice of medicine depends on the malaria parasite species, severity of illness, local drug resistance patterns and individual factors.

• Early diagnosis and prompt treatment are essential.

• Uncomplicated malaria should be treated according to national treatment guidelines.

• Artemisinin-based combination therapy (ACT) is an important treatment approach for *P. falciparum* malaria.

• Severe malaria requires urgent hospital-based management and may require injectable antimalarial therapy.

• Complete treatment is important to cure infection and reduce onward transmission.

Prevention and Control

Prevention focuses on avoiding mosquito bites, controlling mosquito populations and ensuring early detection and treatment.

Personal protection:

• Sleep under insecticide-treated mosquito nets where appropriate.

• Use mosquito repellents.

• Wear clothing that covers exposed skin, especially during mosquito-biting hours.

• Use screens and other measures to reduce mosquito entry into homes.

Vector control:

• Indoor residual spraying (IRS) in appropriate areas.

• Use of long-lasting insecticidal nets (LLINs).

• Identification and management of mosquito breeding habitats.

• Community participation in local vector-control activities.

When to Seek Immediate Medical Attention

• Fever with chills after staying in or travelling to a malaria-prone area

• Confusion, drowsiness or seizures

• Difficulty breathing

• Persistent vomiting or inability to drink fluids

• Dark urine, jaundice or unusual bleeding

• Rapid worsening of symptoms

Malaria: Quick Summary

Cause: Plasmodium parasites
Spread: Bite of an infected female Anopheles mosquito
Incubation: Commonly around 10–15 days
Main symptoms: Fever, chills, headache, sweating and body aches
Severe disease: May cause cerebral malaria, severe anaemia, respiratory distress and organ dysfunction
Diagnosis: Blood smear microscopy and/or rapid diagnostic tests
Treatment: Prompt species- and severity-appropriate antimalarial treatment
Prevention: Mosquito-bite protection, insecticide-treated nets and vector control

References: National Health Portal (NHP), Ministry of Health & Family Welfare, Government of India; National Centre for Vector Borne Diseases Control (NCVBDC), Ministry of Health & Family Welfare, Government of India.

This is an educational study summary and not a substitute for professional medical advice, diagnosis or treatment.